

PDF Generation: Cheat Sheet

Baseline 2026-10-09: iText 9.8.0, JasperReports 7.0.8, PDFBox 3.0.8, EU DSS 6.5, ZXing 3.5.4. Quick reference only: verify against the official documentation and the full pages. Content was generated with AI assistance.

iText 9.8 MODULE MAP

As of 2026-10-09: **itext-core 9.8.0** (modules commons, io, kernel, layout, forms, sign, barcodes, pdfa, pdfua, svg, styled-xml-parser) + **bouncy-castle-adapter** (or the FIPS adapter) for signing and encryption. Java 21 baseline in these pages.

- **Lifecycle:** PdfWriter → PdfDocument → Document (layout) → close. Closing flushes the xref and trailer.
- Reading: PdfReader; stamping or editing: new PdfDocument(reader, writer); incremental: StampingProperties.useAppendMode().
- Merge: PdfMerger; split: PdfSplitter; copy pages with copyPagesTo.

```
try (PdfDocument pdf = new PdfDocument(new PdfWriter(out));
Document doc = new Document(pdf, PageSize.A4)) {
doc.add(new Paragraph("Invoice"));
}
```

LAYOUT, TABLES, EVENTS, FONTS

- Elements: Paragraph, Text, Table, Cell, List, Image, Div, AreaBreak, LineSeparator. Coordinates: origin bottom-left, 1 pt = 1/72 in.
- Tables: UnitValue.createPercentArray widths; header/footer rows; setSkipFirstHeader; Table(n, true) + flush() + complete() for large tables.
- Events: PdfDocumentEvent.START_PAGE / END_PAGE via IEventHandler; "Page X of Y" with a placeholder PdfFormXObject filled at close.
- Custom layout: subclass a renderer (TableRenderer, ParagraphRenderer) and return it from getNextRenderer().
- Fonts: PdfFontFactory.createFont(path, PdfEncodings.IDENTITY_H, EmbeddingStrategy.FORCE_EMBEDDED); embed subset; the 14 standard fonts are WinAnsi only. Complex scripts need pdfCalligraph (commercial).
- HTML: pdfHTML.HtmlConverter.convertToPdf(html2pdf 6.3.4, AGPL or commercial).

PDF/A AND PDF/UA

- PdfADocument(writer, PdfAConformance.PDF_A_2B, new PdfOutputIntent(...)): embed all fonts, sRGB ICC output intent, XMP metadata, no encryption or transparency issues for the level.
- PDF/A-1 (2005, PDF 1.4), -2 (PDF 1.7), -3 (embedded files), -4 (PDF 2.0). Levels b basic, u Unicode, a accessible (tagged).
- PDF/UA-1 (ISO 14289-1) and UA-2 (PDF 2.0): pdf.setTagged(), document language and title, alt text, real header cells (scope), artifacts for decoration.
- Validate with veraPDF and PAC; a validator pass is not proof of usability.

LICENCES AND TOOLS

- **iText Core 9 and pdfHTML, pdfSweep, pdfOCR:** AGPLv3 or commercial.
- **pdfCalligraph, pdfXFA, pdfOptimizer, pdfOffice, pdfRender:** commercial only. AGPL applies to network services too.
- **JasperReports:** LGPL (Professional is separate). **PDFBox:** Apache-2.0. **EU DSS:** LGPL-2.1. **ZXing:** Apache-2.0. **OpenPDF 3:** LGPL or MPL.

JASPERREPORTS 7.0.8 LIFECYCLE

- .jrxml (v7 Jackson JRXML, <element kind="..." />) → JasperCompileManager → JasperReport (.jasper) → JasperFillManager.fillReport(report, params, dataSource) → JasperPrint → exporter.
- **Bands** in print order: title, pageHeader, columnHeader, group header, detail, group footer, columnFooter, pageFooter, lastPageFooter, summary; noData and background extra.
 - **Expressions** (Java): \${name} parameter, \${name} field, \${name} variable (Sum, Count, Average, ..., reset type Report, Page, Column, Group), \${key} resource bundle.
 - **Data sources:** JRBeanCollectionDataSource, JRMapCollectionDataSource, JsonDataSource, JRCsvDataSource, JDBC Connection + SQL query, subreports with their own source.
 - Groups need sorted input (SQL ORDER BY). Crosstab = row groups x column groups x measures. Charts via the chart components (v7 plugin modules).
 - Compile once at startup, fill per request; never let users upload JRXML or scripts (net.sf.jasperreports.compiler.*, expressions run as code).

JASPER PDF EXPORT AND SCALE

```
JRPdfExporter e = new JRPdfExporter();
e.setExporterInput(new SimpleExporterInput(print));
e.setExporterOutput(new SimpleOutputStreamExporterOutput(out));
var c = new SimplePdfExporterConfiguration();
c.setMetadataTitle("Sales");
c.setCompressed(true);
e.setConfiguration(c); e.exportReport();
```

- SimplePdfReportConfiguration: page range, force-line-break; setPdfaConformance(PdfaConformanceEnum...), setTagged; PDF/A-1a, 2a and 3a need tags.
- **Tagged PDF needs Professional** in 7.0.8 (community uses SilentPdfTaggerFactory); the PDF exporter uses OpenPDF (jasperreports-pdf).
- Large reports: JRFileVirtualizer / JRSwapFileVirtualizer, and stream the result to the response.

MIGRATING FROM THE BOOKS

Old (iText 2 and 5)	Use now
com.lowagie.*	com.itextpdf.* (or org.openpdf)
PdfWriter.getInstance	new PdfDocument(new PdfWriter)
PdfPTable, Phrase	Table, Text, Paragraph
PdfStamper, AcroFields	PdfDocument(reader, writer), PdfAcroForm
PdfPKCS7 flow	PdfSigner + SignerProperties
JR DTD / XSD JRXML	v7 Jackson JRXML (kind)
JRExporterParameter	Simple*Configuration classes

Full list O1 to O28 and examples: see the page *What's Changed and Migration*.

PADES LEVELS

- | Level | Adds |
|-------|--|
| B-B | CMS signature + signing certificate |
| B-T | signature timestamp (RFC 3161 TSA) |
| B-LT | validation data in a DSS dictionary (certs, OCSP, CRL) |
| B-LTA | document timestamp + archive refresh |
- **iText:** new PdfSigner(reader, out, new StampingProperties().useAppendMode()), SignerProperties (field, reason, location, appearance), signer.signDetached(digest, signature, chain, crl, ocsp, tsa, 0, CryptoStandard.CADES). LTV via LtvVerification.
 - **PDFBox:** pdDoc.addSignature(sig, signatureInterface, options) + saveIncremental; you write the CMS, no PAdES levels built in.
 - **EU DSS 6.5:** PAdESService.signDocument with PAdESSignatureParameters and SignatureLevel.PAdES_BASELINE_B / _T / _LT / _LTA; extendDocument upgrades B to LT or LTA.
 - Signature /ByteRange covers all bytes but the /Contents hole; any later edit must be an incremental update that keeps it valid.

ACROFORM FIELDS

- iText 9** PdfFormCreator.getAcroForm(pdf, true); the six builders replaced the static PdfFormField.createXxx factories (8.0):
- TextFormFieldBuilder, ChoiceFormFieldBuilder (combo, list), CheckBoxFormFieldBuilder, RadioFormFieldBuilder (group + buttons), PushButtonFormFieldBuilder, SignatureFormFieldBuilder.
- ```
PdfTextFormField f = new TextFormFieldBuilder(pdf, "customer").setWidgetRectangle(r).createText();
form.addField(f);
form.flattenFields();
```
- Always set an explicit font and appearance streams; avoid NeedAppearances and XFA (deprecated in PDF 2.0, forbidden in PDF/A, XFA flatten = pdfXFA, commercial).
  - **JasperReports** sets fields with element properties net.sf.jasperreports.export.pdf.field.type = Text, Combo, List, Check or Radio, plus .name, .value, .choices, .choice.separators, .combo.edit, .check.type (since 6.12.0).
  - **PDFBox:** PDAcroForm, PDTextFormField, PDCheckBox, PDRadioButton, PDComboBox, PDListBox, PDSignatureField; acroForm.flatten().

## BARCODES AND QR CODES

- **iText:** BarcodeQRCode, Barcode128, BarcodeEAN, Barcode39, BarcodePDF417, BarcodeDataMatrix; draw with createFormXObject(pdf) into an Image (vector).
- **ZXing 3.5.4:** QRCodeWriter / MultiFormatWriter → BitMatrix; hints ERROR\_CORRECTION, MARGIN, CHARACTER\_SET.
- **JasperReports:** <element kind="component"> Barcode4j, kind="barbecue", QR as an image expression from ZXing.

| QR EC | Recovers |
|-------|----------|
| L     | ~7 %     |
| M     | ~15 %    |
| Q     | ~25 %    |
| H     | ~30 %    |

- Quiet zone of 4 modules, print at least ~2 cm square for invoices; higher EC = bigger symbol. VeriFactu QR uses level M and the AEAT URL format; use the pre-production host in tests.